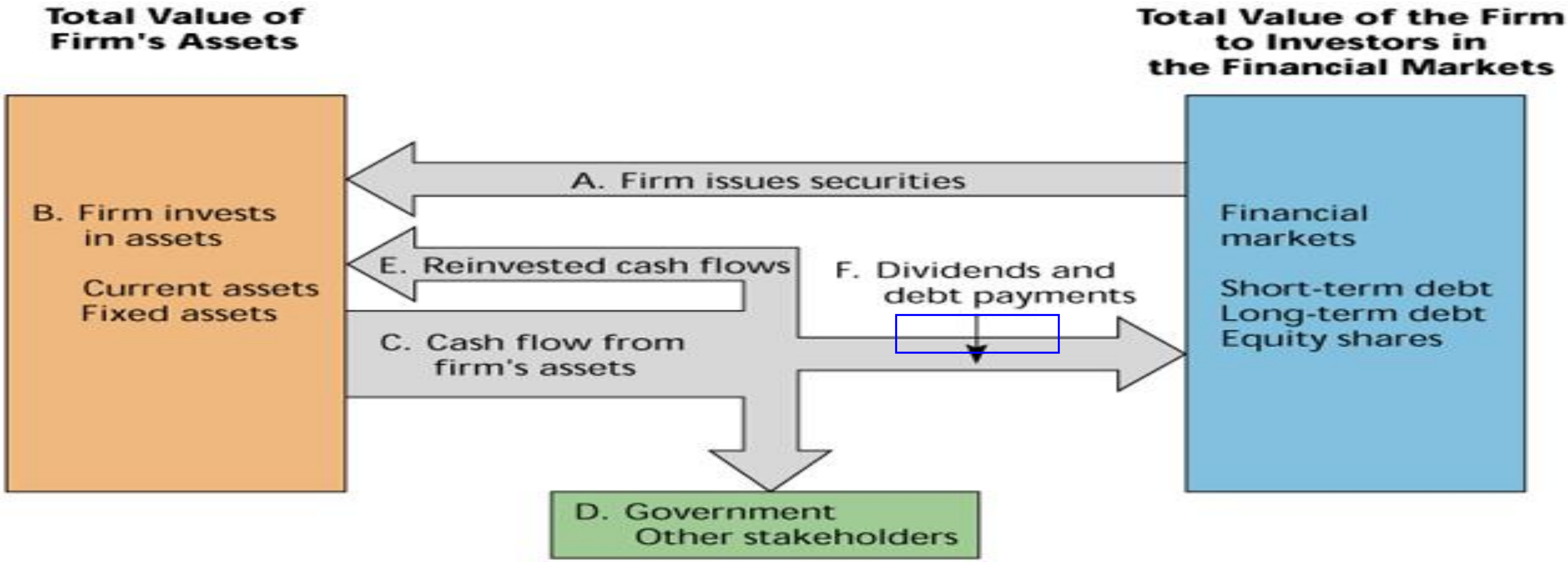
Financial markets

2025-1

Relationship between a Firm and the Financial Market



A. Firm issues securities to raise cash.
B. Firm invests in assets.
C. Firm's operations generate cash flow.
D. Cash is paid to government as taxes. Other stakeholders may receive cash.
E. Reinvested cash flows are plowed back into firm.
F. Cash is paid out to investors in the form of interest and dividends.

Financial Markets and the Economy

- Information Role
- Consumption Timing
- Allocation of Risk
- Separation of Ownership and Management
  - Agency Problem: 확증비용(bonding cost), 감시비용(monitoring cost), 잔여손실(residual cost)

Asset classes

- Money markets vs. capital markets
- Types of money market instruments
- Capital market securities:
  - Bonds
  - Equity
  - Derivatives

The Money Market

- Subsector of the fixed-income market: Securities are short-term, liquid, low risk, and often have large denominations

- Money market mutual funds allow individuals to access the money market

- https://www.investopedia.com/terms/m/moneymarket.asp

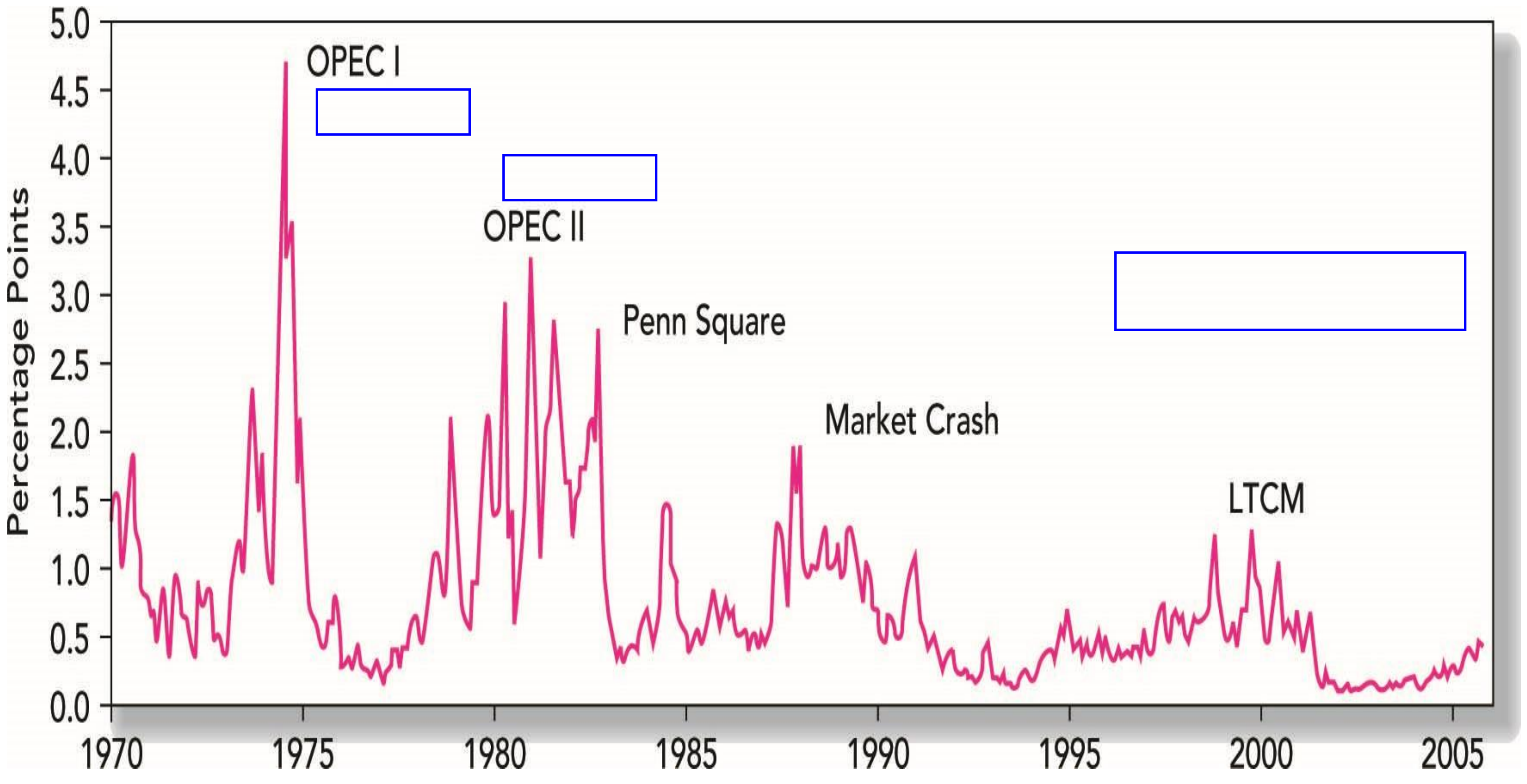
Money Market Securities

- Treasury bills: short-term (ranging from a few days to 52 weeks) debt of U.S. government
  - typically sold at a discount from the par amount (par amount is also called face value)
- Certificates of deposit (CD): savings account that holds a fixed amount of money for a fixed period of time, such as one, three, or six months, and in exchange, the issuing bank pays interest
- Commercial paper: short-term, unsecured debt of a company

Money Market Securities

- Bankers' Acceptances: an order to a bank by a bank's cus tomer to pay a sum of money on a future date
- Eurodollars: dollar-denominated time deposits in banks outside the U.S.
- Repos and reverses: short-term loan backed by govern-m ent securities
- Fed funds: very short-term loans between banks

Spread between 3-month CDs and Treasury Bill Rates



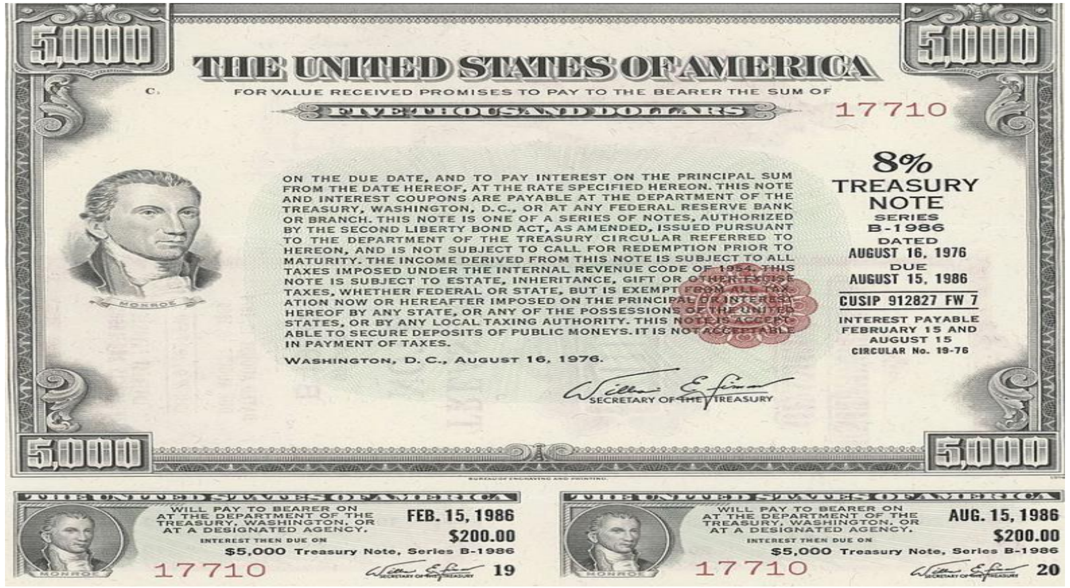
https://fred.stlouisfed.org/series/TY3MCD

The Bond Market

- Treasury Notes (1-10yrs) and Bonds (10+ yrs)
- Treasury Inflation-Protected Securities (TIPS)
- Federal Agency Debt
- International Bonds (Eurodollar bond)
- Municipal Bonds
- Corporate Bonds
- Mortgage-Backed Securities

Bond Market Securities

- Treasury Notes and Bonds
  - Maturities
    - Notes – Maturities up to 10 years
    - Bonds – Maturities from 10 to 30 yea rs
  - Par Value - $1,000~1,000,000
  - Interest paid semiannually
  - Quotes – Percentage of par



Bond Market Securities

- Treasury Inflation-Protected Securities (TIPS)
  - Pay interest based on face value adjusted to Consumer Pric e Index; at maturity face value is guaranteed.
- Federal Agency Debt
  - Debt of mortgage-related agencies such as Fannie Mae and Freddie Mac
- International Bonds
  - Eurobonds
  - Yankee bond, Samurai bond: foreign bonds

Bond Market Securities

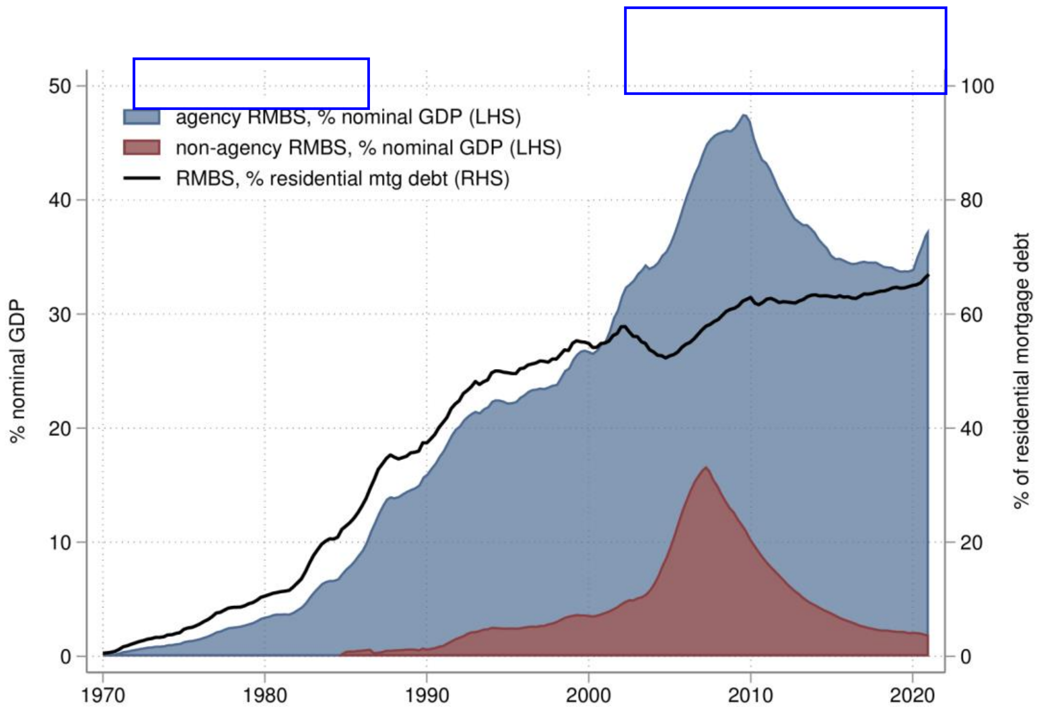
- Municipal Bonds
  - Issued by state and local governments
  - Interest is exempt from federal income tax and sometimes from state an d local tax
  - Types
    - General obligation bonds: Backed by taxing power of issuer
    - Revenue bonds: backed by project's revenues or by the municip al agency operating the project.

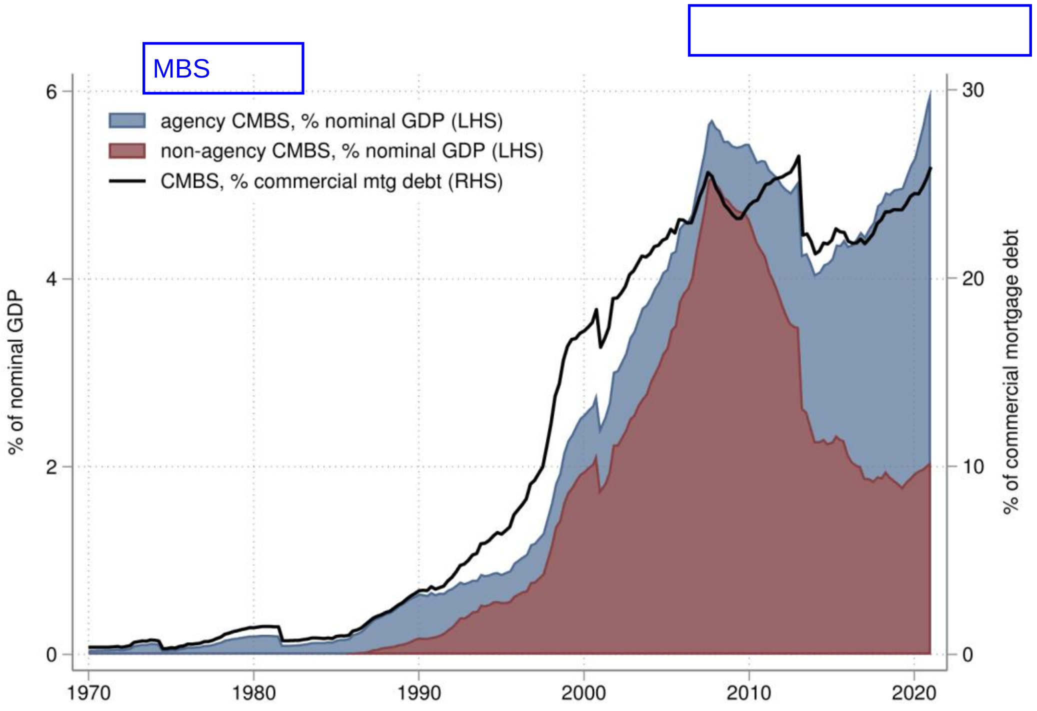
Bond Market Securities

- Corporate Bonds
  - Issued by private firms
  - Semi-annual interest payments
  - Subject to larger default risk than government securities
  - Options in corporate bonds
    - Callable
    - Convertible

Bond Market Securities

- Mortgage-Backed Securities (MBS)
  - Proportional ownership of a mortgage pool or a specified obligation secured by a pool
  - Produced by securitizing mortgages
    - Mortgage-backed securities are called *pass-throughs* becau se the cash flows produced by homeowners paying off thei r mort-gages are passed through to investors
  - Most were issued by Fannie Mae and Freddie Mac


agency RMBS, % nominal GDP (LHS)
non-agency RMBS, % nominal GDP (LHS)
RMBS, % residential mtg debt (RHS)
% nominal GDP
% of residential mortgage debt
0
10
20
30
40
50
0
20
40
60
80
100
1970
1980
1990
2000
2010
2020


MBS
agency CMBS, % nominal GDP (LHS)
non-agency CMBS, % nominal GDP (LHS)
CMBS, % commercial mtg debt (RHS)
% of nominal GDP
% of commercial mortgage debt
0
2
4
6
0
10
20
30
1970
1980
1990
2000
2010
2020

Equity Securities

- Common stock: Ownership
  - Residual claim
  - Limited liability
- Preferred stock: Perpetuity
  - Fixed dividends
  - Priority over common
  - Tax treatment
- American Depository Receipts
  - Certificates traded in U.S. markets that represen t ownership in shares of a foreign company

Stock Market Indexes

- Dow Jones Industrial Average (DJIA)
  - Includes 30 large blue-chip corporations
  - Computed since 1896
  - Price-weighted average

Price-Weighted Average

Portfolio: Initial value $25 + $100 = $125

Final value $30 + $90 = $120

Percentage change in portfolio value

= 5/125 = -.04 = -4%

Index: Initial index value (25+100)/2 = 62.5

Final index value (30 + 90)/2 = 60

Percentage change in index -2.5/62.5

= -.04 = -4%

Stock Market Indexes

- Standard & Poor's 500 S&P 500
  - Broadly based index of 500 firms
  - Market-value-weighted index
- Investors can base their portfolios on an index
  - Buy an index mutual fund
  - Buy exchange traded funds (ETFs)

Other Indexes

U.S.

- NYSE Composite
- NASDAQ Composite
- Wilshire 5000

Non U.S.

- Nikkei (Japan)
- FTSE (U.K.; “footsie”)
- DAX (Germany)
- Hang Seng (Hong Kong)
- TSX (Canada)
- KOSPI (Korea)

Performance of U.S. Major Stock Indexes

Major Stock Indexes

	DAILY					52-WEEK			YTD
Dow Jones Averages	HIGH	LOW	CLOSE	NET CHG	%CHG	HIGH	LOW	%CHG	%CHG
30 Industrials	10760.57	10672.60	10709.74	−2.48	−0.02	11043.44	10012.36	+ 2.01	−0.07
20 Transportations	4298.16	4223.18	4240.81	−34.21	−0.80	4275.02	3382.89	+19.88	+1.07
15 Utilities	427.50	416.70	418.62	−6.40	−1.51	437.63	337.39	+24.08	+3.33
65 Composite	3711.35	3660.44	3674.60	−22.01	−0.60	3711.52	3211.74	+11.87	+1.00
Dow Jones Indexes									
Wilshire5000	12847.74	12722.78	12772.31	−29.61	−0.23	13011.23	11217.81	+10.52	+2.03
Wilshire 2500	3041.31	3010.60	3022.79	−7.31	−0.24	3083.08	2662.75	+10.39	+1.94
DJ STOXX 50	3362.23	3325.29	3362.19	+35.66	+1.07	3444.10	2803.26	+19.92	+0.39
Nasdaq Stock Market									
Nasdaq Comp	2275.30	2253.06	2260.65	−4.60	−0.20	2331.36	1904.18	+10.49	+2.51
Nasdaq 100	1694.54	1670.33	1677.00	−9.22	−0.55	1758.24	1406.85	+11.13	+1.93
Biotech	815.23	807.25	808.70	−1.46	−0.18	834.42	641.35	+11.36	+2.33
Computer	1015.13	1000.15	1005.03	−2.56	−0.25	1066.53	845.22	+9.88	+1.33
Standard & Poor's Indexes									
500Index	1271.87	1259.42	1264.68	−2.18	−0.17	1294.18	1137.50	+7.72	+1.31
MidCap 400	771.06	759.94	762.96	−4.99	−0.65	772.41	627.38	+19.65	+3.38
SmallCap 600	371.71	367.99	369.89	−0.62	−0.17	370.51	301.06	+16.99	+5.48
SuperComp 1500	288.79	285.85	287.05	−0.61	−0.21	293.15	255.39	+9.06	+1.66
New York Stock Exchange and Others									
NYSE Comp	8009.18	7943.17	7969.49	−3.53	−0.04	8063.51	6935.31	+13.10	+2.78
NYSE Financial	8123.88	8079.87	8107.37	+27.45	+0.34	8299.27	6905.21	+11.97	+1.38
Russell 2000	720.30	713.13	716.45	−1.57	−0.22	718.02	575.02	+16.20	+6.42

Source: Reuters

Examples of International Indexes

- Country Indexes

- Nikkei 225 (Tokyo stock exchange listed 225 firms)
- FTSE 100 (London stock exchange listed 100 firms)
- DAX 30 (Frankfurt stock exchange listed 30 firms)

- Region Indexes
  - EAFE: Europe, Australia, Far East composite index maintained by MSCI

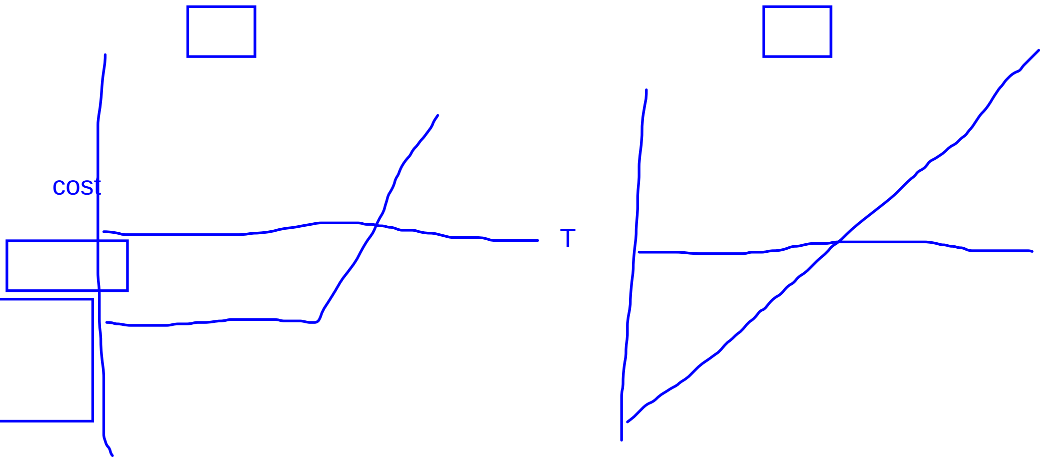
Construction of Indexes

- How are stocks weighted?
  - Price weighted (DJIA)
  - Market-value weighted (S&P500, NASDAQ)
  - Equally weighted (Value Line Index)

Derivatives Markets

- A derivative is a security that gets its value from the values of another asset, such as commodity prices, bond and stock prices, or market index values.

Derivatives Markets



- Options
  - *Call*: **Right to buy** underlying asset at the strike or exercise price
    - Value of calls decreases as strike price increases
  - *Put*: **Right to sell** underlying asset at the strike or exercise price
    - Value of puts increase with strike price
  - Value of both calls and puts increases with time until expiration

Derivatives Markets

- **Futures** Contracts
  - An agreement made today regarding the **delivery** of an asset (or in some cases, its cash value) at a specified delivery or maturity date for an agreed-upon price, called the futures price, to be paid at contract maturity
  - ***Long*** *position*: Take delivery at maturity
  - ***Short*** *position*: Make delivery at maturity

Comparison

Options

- **Right, not obligation** to buy or sell; option is exercised when it is profitable

- Options are traded with the premium, the price of the option itself.

Futures

- **Obliged to make or take delivery**; long positioner must buy at the futures price, short positioner must sell at futures price

- Futures contracts with low cost (margin)
- Margin: margin call

코스피200 선물증거금은 코스피200선물지수 × 거래승수(25만원) × 증거금률 로 계산됩니다.

예를 들어, 코스피선물지수가 335 포인트라고 가정한다면

위탁증거금 = 335 × 25만원 × 7.5% = 628.125만원

유지증거금 335 × 25만원 × 5% = 418.75만원 입니다.